Common Problems and Solutions

I. After starting the program, the camera has no image and the robotic arm cannot be controlled

Answer: The most likely cause is loose wiring or the system exiting abnormally, which leads to resource contention. Solution: Exit all programs, then unplug and re-plug the USB camera and the Micro-USB data cable.

II. The grabbing/placing positions and angles are incorrect, and the block is not gripped tightly

Answer: You can use the Arm Calibration Guide in the mobile app to calibrate the servos.

III. Color recognition performance is poor

Answer: You can perform HSV color calibration, record the HSV values of the correct color, and write them into the color recognition program. For details, refer to the AI Vision Course, "3. Color Calibration."

IV. Jupyter notebook case reports errors or stalls

Answer: Restart the Jupyter Kernel. If issues persist, restart the virtual machine.

V. The servo behaves abnormally, but can be controlled individually

Answer: Check whether the power supply is normal.

VI. Garbage classification cannot be recognized properly

Answer: Garbage classification recognition requires placing a small block at a fixed position on the map, and the side with characters should face the robotic arm.

VII. The "luring out the snake" effect is relatively hard to identify

Answer: Use color blocks to repeatedly move the arm back and forth, then slowly move away to trigger the effect. This case requires multiple attempts.

VIII. In cases using gesture recognition, the robotic arm cannot recognize gestures

Answer: Generally, this is because the gesture recognition API usage limit has been exhausted. You need to apply for an account and replace the keys in the program. The specific steps can refer to the AI Vision Course, in the "Gesture Recognition Fixed Actions" case. After obtaining the account, modify app_id, api_key, and secret_key in the corresponding source code of the case.

APP Case:/home/yahboom/colcon_ws/src/dofbot_gesture/gesture_action.py

Gesture Recognition: Dofbot/6.AI_Visual/4.Gesture recognition.ipynb

Gesture Stacking: Dofbot/6.AI_Visual/2.gesture_stack.ipynb

Gesture Recognition Fixed Action:/home/yahboom/Dofbot/6.AI_Visual/1.gesture_action.ipynb

IX. The robotic arm starts flailing randomly or movement directions are undefined

Answer: An unestimated flailing or undefined direction issue may be caused by the servo IDs being cleared. You can go to the "Single Servo Control" case in the "Fundamental Control Course" to test, and verify whether each servo can be controlled normally.

X. Running the large program and JupyterLab case concurrently may cause errors

Answer: Yes, conflicts may occur; running JupyterLab related cases requires closing the large program first.

XI. After upper computer color calibration, the robotic arm cannot return to center

Answer: After calibration, remember to click Cancel; afterwards you can control the robotic arm normally.